Tianyu Xiong

Ph.D. Candidate / 3D Computer Vision

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six-bit-tx.github.io/research/

RESEARCH PROFILE

Ph.D. candidate in Computer Science and Technology at Northwestern Polytechnical University. My publications cover 3D reconstruction, Gaussian splatting, watermarking and machine learning.

EDUCATION

Northwestern Polytechnical University 2024–present

Ph.D. in Computer Science and Technology Xi'an, China

Supervisors: Yanning Zhang and Yangjia Qi.

University of Toronto 2018–2021

Honours Bachelor of Science St. George Campus, Toronto, Canada

Computer Science Specialist.

PUBLICATIONS

Queryable Gaussian View Index: Text Retrieval and Selective Camera Localization

Tianyu Xiong, Rui Li, Jiaqi Yang.

Manuscript, 2026. Submitted to arXiv.

SalientGS: Unified SfM-to-3DGS with Importance-Guided MCMC Gaussian Allocation

Tianyu Xiong, Rui Li, Suning Ge, Jiaqi Yang.

ACM Multimedia 2026. Accepted. arXiv:2607.11285

CAGE-LIVO: Cooperative Air-Ground Gaussian-Enhanced LiDAR-Inertial-Visual Odometry

Tianyu Xiong et al. (first author).

Chinese Journal of Aeronautics. Accepted.

VoMarkSplat: Robust watermarking for 3D Gaussian splatting with patch and multi-convolutional voting

Tianyu Xiong, Linjie Li, Rui Li, Jiaqi Yang, Yanning Zhang.

Pattern Recognition Letters, 203:8–14, May 2026. DOI

GloSplat: Joint Pose-Appearance Optimization for Faster and More Accurate 3D Reconstruction

Tianyu Xiong, Rui Li, Linjie Li, Jiaqi Yang.

Preprint, 2026. arXiv:2603.04847

Large scale air pollution prediction with deep convolutional networks

Gao Huang, Chunjiang Ge, **Tianyu Xiong**, Shiji Song, Le Yang, Baoxian Liu, Wenjun Yin, Cheng Wu.

Science China Information Sciences, 64(9), article 192107, 2021. DOI

The effect of similarity on the evolution of fairness in the ultimatum game

Yakun Zhao, **Tianyu Xiong**, Lei Zheng, Yumeng Li, Xiaojie Chen.

Chaos, Solitons & Fractals, 131, article 109494, February 2020. DOI

TECHNICAL FOCUS & LANGUAGE

Methods: Gaussian splatting, differentiable rendering, multi-view geometry, neural-network learning and CLIP-based vision–language models.

English: Native-level proficiency.

Research & experience

Tianyu Xiong / 3D computer vision and machine learning

SELECTED COMPLETED RESEARCH

SalientGS Fast reconstruction and capacity allocation

Developed a unified reconstruction pipeline that jointly refines camera poses and Gaussian appearance. Importance-guided MCMC allocation directs a fixed Gaussian budget toward persistently underfit regions.

VoMarkSplat Watermarking of 3D Gaussian assets

Developed robust watermark recovery for Gaussian scene assets through spatial-patch voting, multi-convolutional voting and a two-stage training curriculum.

EARLIER RESEARCH EXPERIENCE

The Hong Kong Polytechnic University May 2022–May 2024
Department of Computing Hong Kong, China

Advisor: Lei Zhang; collaborative projects with Hongwei Yong.

- Built a CLIP-based prototype for image cropping guided by content and aesthetic instructions.
- Studied feature subsampling during the backward pass of convolutional neural networks and parallelizable feedback mechanisms based on Direct Feedback Alignment.

Tsinghua University May–August 2019
Department of Automation Beijing, China

Advisor: Gao Huang. Contributed to spatiotemporal air-pollution prediction with deep convolutional networks, leading to the 2021 Science China Information Sciences publication.

Beihang University May–August 2018
School of Electronic and Information Engineering Beijing, China

Advisor: Xiantong Zhen. Implemented CliqueNet for MURA radiographic image classification.

ENTREPRENEURSHIP

Halfman Tech · Co-founder May–September 2024
Beijing, China

Co-founded an early-stage technology company; operations ended in September 2024.